



DEALER BEST PRACTICE SERIES

Engine Bay Buggy Steps

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

An increased number of technician incidents have been caused by improper working platforms in Off Highway Truck (OHT) engine bays. The engine components and accompanying supports provide an unsuitable surface for supporting technician footing during servicing. The lack of ergonomic working area has caused increased technician fatigue and incidents. On some occasions, technicians using these as foot holds have caused damage to electrical and hydraulic components.

This best practice details the addition of foldaway “buggy steps” that can be used by technicians during servicing to provide adequate working platforms. The installation of well-positioned steps should also reduce or eliminate the need to use ladders and platforms from ground level for adequate support.

2.0 Best Practice Description

A current practice for many technicians is to complete OHT engine servicing using work platforms or ladders for support from ground level. Shifting these platforms is often a timely exercise and often these platforms are blocked by engine cross members or strut mounts. Subsequently it is a common scene where technicians utilize the chassis and other components in the engine bay as foot holds while completing servicing.

Support steps can be fitted in strategic locations onto the chassis rail where sufficient clearance is available. As shown in Figure 1, the supports should be placed in a conspicuous area while having little to no effect on other systems located in the engine bay. If the mounts are to be welded on any areas other than shown in this Best Practice, it should be verified that the position will not affect the integrity of the component the mount is welded to. The Dealers Solutions Network (DSN) can be contacted to assure the location of the mount will not affect the reliability of the component to be welded on. Hand holds also should be present to assist in the use of the steps.

After servicing, the supports can be folded away to maintain space beside the engine. The supports should also be installed using the four fastening points for easy removal if required.

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Fig. 1 Buggy step installed onto weld-plate

Different sized supports can be used where clearance is an issue. It is suggested that the supports should be of similar design to those shown in Figure 3 as well as having a load rating and an adequate slip resistant surface.



Fig. 2 Different sized steps installed on strut mount and frame rail

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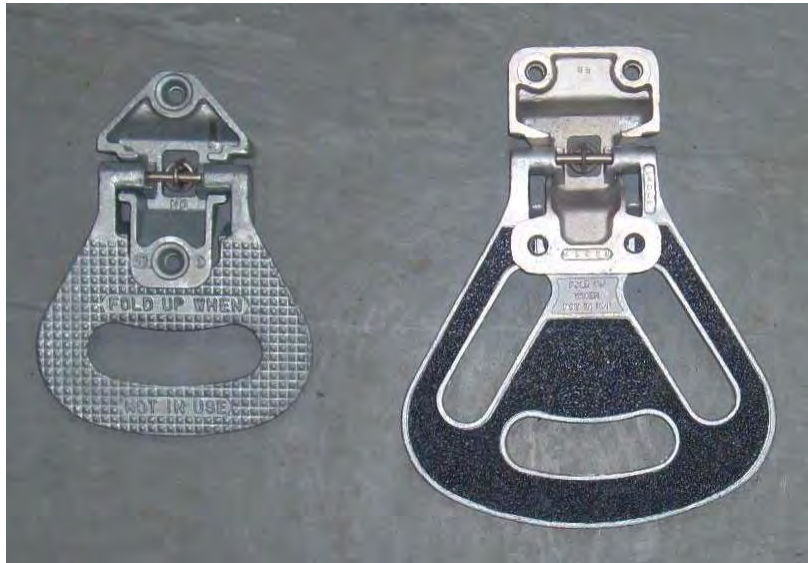


Fig. 3 Example of foot hold products

3.0 Implementation Steps

1. Buggy steps shown in the best practice have been installed on a 793C OHT, but installation can be replicated in OHT's ranging from 777 – 797.
2. The foot-hold products shown above can be procured from the Caterpillar parts system by referencing Part Number 153-3861.
3. Weld plates can be fabricated from 12mm Grade 250 mild steel plate with M8 threaded holes to suit foot hold anchor points.
4. 25mm M8x1.25 fasteners (8T-4908) and flat washers (9M-1974) to suit should be used to affix foot holds to threaded weld plates.
5. Weld plate can be welded to chassis frame with a minimum of 6" from structural weld seams or joins. **Weld seams must run parallel to the chassis rail only to mitigate high stresses;** Figure 1 shows weld plate fixed to the chassis with seams on both top and bottom edges. All weld quality shall conform to American National Standards Institute (ANSI)/American Welding Society (AWS) D14.3. Consult DSN if welding on a location other than what is shown in this Best Practice.

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4.0 Benefits

1. 450kg load rated, slip resistant “buggy steps” provide a safer platform for technician footing.
2. Steps allow mechanics to more safely move about in engine bays.
3. Mechanics can position themselves more comfortably in engine bays for more ergonomic working conditions.
4. Steps provide for less cramped working conditions.
5. Spring-loaded steps fold away flat against chassis rail when not in use.
6. Minimized usage of ladders to access engine bay area.

5.0 Resources Required

Step Part Number (153-3861).
25mm M8x1.25 fasteners (8T-4908).
Flat Washers (9M-1974).
Fabrication of threaded weld plate required.
SMA Welding required on truck body.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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